Three-level hierarchical POMDP: an executed test of the N-level template I

The 2-level hierarchical POMDP (sec. 13.9) couples location inference to a single global context. The N-level architecture (`fedference.pomdp.build_nlevel_world`) provides a parameterized stack of levels; the canonical 3-level example couples location (L1; 9 states) to a context variable (L2; 2 states: `quiet` / `alert`) and further to a meta-context variable (L3; 2 states: `low_threat` / `high_threat`) that gates the L2 prior.

Three-level hierarchical POMDP: an executed test of the N-level template I

$$\tilde{D}_{\mathsf{L2}} = \sum_k q_{\mathsf{L3}}[k]\, p_{\mathsf{L2|L3}}[k] \tag{1}$$

$$\tilde{D}_{\mathsf{L1}} = \sum_c q_{\mathsf{L2}}[c]\, p_{\mathsf{L1|L2}}[c] \tag{2}$$

The inference algorithm (`fedference.pomdp.nlevel_infer`) performs 4 passes of top-down / bottom-up alternating minimization: the top-down pass propagates empirical priors from L3 $\rightarrow$ L2 $\rightarrow$ L1 via eq. 1 and eq. 2; the bottom-up pass updates each level's belief from the marginal evidence contributed by the level below.

We compare two conditions over 960 seeded trials with 4 agents at sensor acuity 0.85:

Three-level hierarchical POMDP: an executed test of the N-level template I

- **Flat** — agents ignore all hierarchy and infer location under a uniform prior;
- **3-level** — agents run 4 alternating-minimization iterations across all three levels before federating.

The measured location accuracies are 0.984 (flat) and 0.966 (3-level), a gap of -0.019. Across 128 independent seeds the 3-level location accuracy is 0.976 (SD 0.005; 95 % CI 0.976–0.977) versus flat 0.981 (95 % CI 0.980–0.981), a mean accuracy gap of -0.004 (95 % CI -0.005—0.003; Wilcoxon signed-rank $p = 0.0000$, effect size $r = 0.724$, medium; the location gap over the flat baseline is not statistically significant at this seed count). The 3-level condition additionally reports context accuracy 0.697 and meta-context accuracy 0.547. Against the two-state chance baseline of 0.5, location is recovered and the intermediate context latent is resolved well above chance, but the meta-context latent is only marginally above chance — the weakest of the three levels — and is therefore *not* convincingly recovered here. The study thus demonstrates that the generic N-level alternating-minimization runs and federates end-to-end and recovers the fastest (location) and intermediate (context) latents; full recovery of the slowest (meta-context) level is left open. The full figure comparing 2-level and 3-level belief dynamics is fig. 25. The declarative layer specification used by the generic constructor is documented in the supplement (sec. 13.12). For the effect of acuity and colony size on these results, see sec. 13.13.